

Statistical Analysis Report – Use Case Diagram Compilation

Hypotheses

H_0 : The proportion of diagrams that compile correctly is equal between the models ($p_A = p_B$).
 H_A : There is a difference in the proportion of diagrams that compile between the models ($p_A \neq p_B$).

Descriptive Statistics

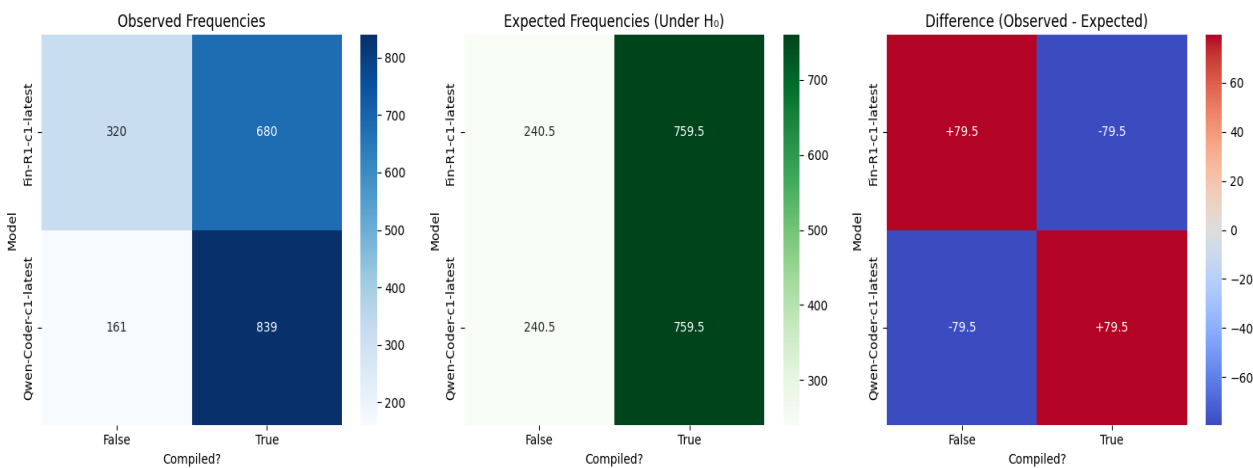
Total samples: 2000
Compiled diagrams: 1519
Non-compiled diagrams: 481

Model	Compiled	Total	Proportion
Fin-R1-c1-latest	680	1000	0.6800
Qwen-Coder-c1-latest	839	1000	0.8390

Z-Test Result

Z statistic: -8.3188
p-value: 8.8858e-17
Conclusion: There is a statistically significant difference between the models ($p < 0.05$).

Visual Analysis of Frequencies



Interpretation of Visual Frequencies

The heatmaps above illustrate the observed frequencies of compilable diagrams (left), the expected frequencies under the assumption of independence (center), and the difference between observed and expected values (right).
Key observations:

- The specialized model (Fin-R1-c1-latest) shows a lower count of successful compilations than expected, suggesting a possible negative impact from specialization.
- The non-specialized model (Qwen-Coder-c1-latest) exceeds its expected successful compilations, reinforcing its superior performance in syntactical correctness.
- The difference map confirms that the deviations are substantial and systematic, not due to random fluctuation.

Integrated Conclusion

The non-specialized model (Qwen-Coder-c1-latest) compiled more diagrams correctly than the specialized one (Fin-R1-c1-latest).

This outcome contradicts the initial hypothesis H■, suggesting that the domain-specific adaptation may have introduced constraints or noise reducing syntactic validity.